

An introduction to Bayesian modelling with brms and Stan

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<https://frodriguezsanchez.net>

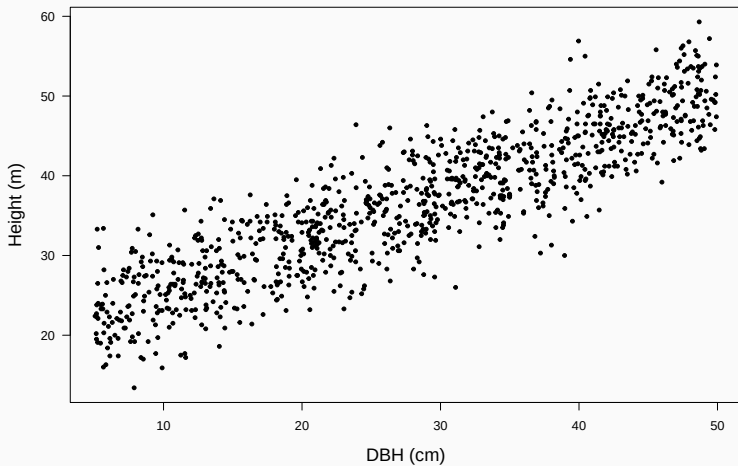
Our dataset: tree heights and DBH

- One species
- 10 plots
- 1000 trees
- Number of trees per plot ranging from 4 to 392

```
trees <- read.csv('data/trees.csv')
```

	site	dbh	height
Min.	: 1.0	Min. : 5.06	Min. :13.40
1st Qu.:	1.0	1st Qu.:17.69	1st Qu.:29.68
Median	: 2.0	Median :28.62	Median :36.55
Mean	: 2.7	Mean :27.88	Mean :36.51
3rd Qu.:	4.0	3rd Qu.:38.97	3rd Qu.:43.33
Max.	:10.0	Max. :49.92	Max. :59.30

What's the relationship between DBH and height?



First step: linear regression (lm)

```
simple.lm <- lm(height ~ dbh, data = trees)
```

Call:

```
lm(formula = height ~ dbh, data = trees)
```

Residuals:

Min	1Q	Median	3Q	Max
-13.3270	-2.8978	0.1057	2.7924	12.9511

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	19.33920	0.31064	62.26	<2e-16 ***
dbh	0.61570	0.01013	60.79	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.093 on 998 degrees of freedom

Multiple R-squared: 0.7874, Adjusted R-squared: 0.7871

F-statistic: 3695 on 1 and 998 DF, p-value: < 2.2e-16

Center continuous variables

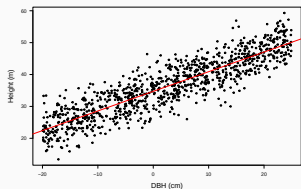
```
summary(trees$dbh)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
5.06	17.69	28.62	27.88	38.97	49.92

```
trees$dbh.c <- trees$dbh - 25
```

So, all parameters will be referred to a 25 cm DBH tree.

Linear regression with centred DBH



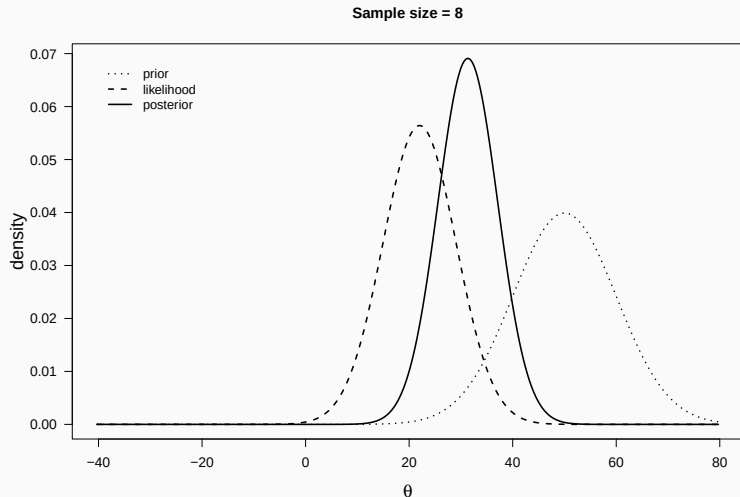
```
lm(formula = height ~ dbh.c, data = tree)
      coef.est coef.se
(Intercept)  34.73    0.13
dbh.c         0.62    0.01
---
n = 1000, k = 2
residual sd = 4.09, R-Squared = 0.79
```

Let's make it Bayesian

Bayesian inference: prior, posterior, and likelihood

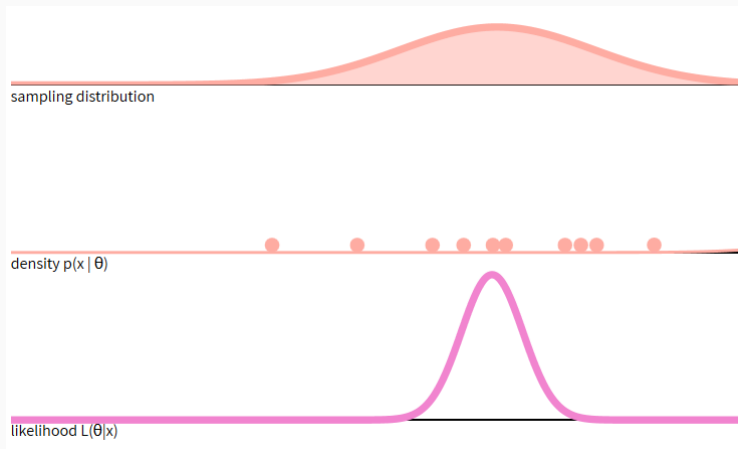
$$P(\text{Unknown}|\text{Data}) \propto P(\text{Data}|\text{Unknown}) \times P(\text{Unknown})$$

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$



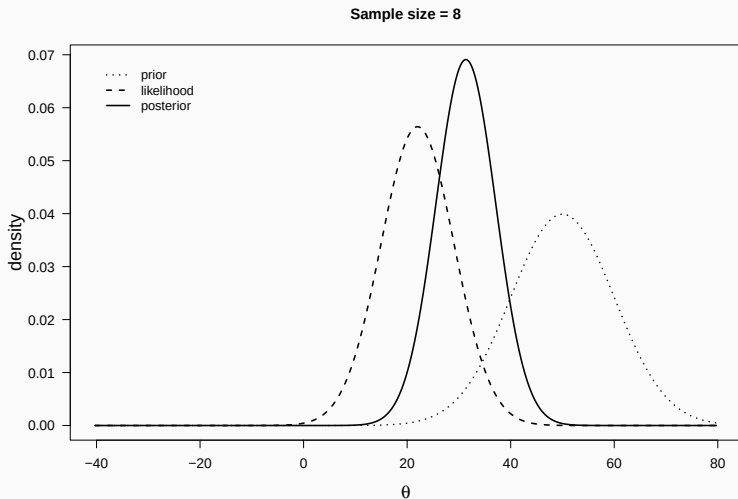
What is the likelihood?

$$L(\theta|x) = P(x|\theta)$$

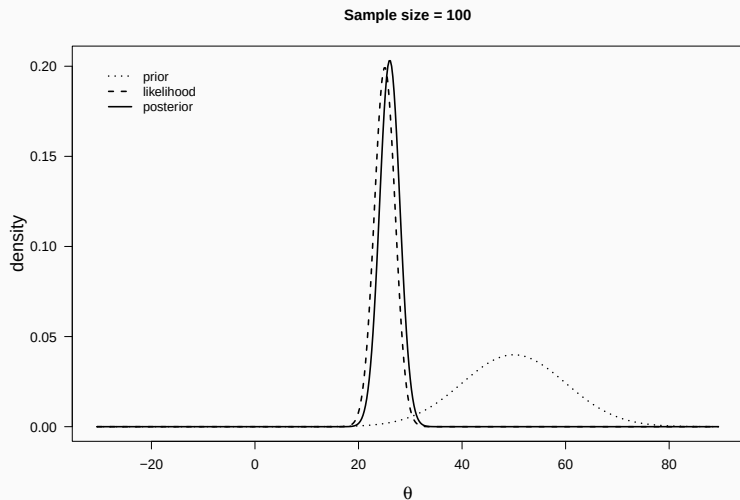


<https://seeing-theory.brown.edu/bayesian-inference/index.html>

Bayesian inference: prior and likelihood produce posterior



With increasing sample size, likelihood dominates prior



- Integrate information (**prior**)

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- Large dataset -> prior effect **diminishes**
- **Uncertainty / Propagate errors**

$$y_i \sim N(\mu_i, \sigma^2)$$

$$\mu_i = \alpha + \beta x_i$$

In this case:

$$\text{Height}_i \sim N(\mu_i, \sigma^2)$$

$$\mu_i = \alpha + \beta \text{DBH}_i$$

α : expected height when DBH = 25 cm

β : how much height increases with every unit increase of DBH

```
library('brms')  
  
height.formu <- brmsformula(height ~ dbh.c)
```

We must define **prior distributions** for every parameter

```
get_prior(height.formu, data = trees)
```

	prior	class	coef	group	resp	dpar	nlpar	lb	ub
	(flat)	b							
	(flat)	b	dbh.c						
student_t(3, 36.5, 10.2)	Intercept								
student_t(3, 0, 10.2)	sigma							0	
source									
default									
(vectorized)									
default									
default									

Choosing priors

Avoid 'non-informative' priors

Use *weakly informative* (e.g. relatively wide Normal or t-student distributions)
or *strongly informative* priors based on previous knowledge and common sense.

Some tips for setting priors:

- <https://github.com/stan-dev/stan/wiki/Prior-Choice-Recommendations>

Run **prior predictive checks** (just priors, no data)

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- [Priors chapter](#) in The BUGS book

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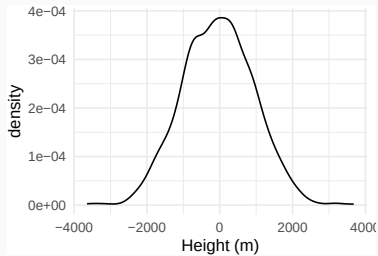
- <https://github.com/stan-dev/stan/wiki/Prior-Choice-Recommendations>
- Priors chapter in The BUGS book
- <https://doi.org/10.1111/oik.05985>

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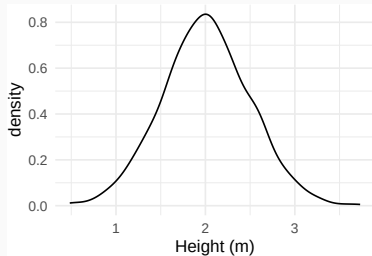
<https://distribution-explorer.github.io/>

Example: estimating people height across countries

Unreasonable prior



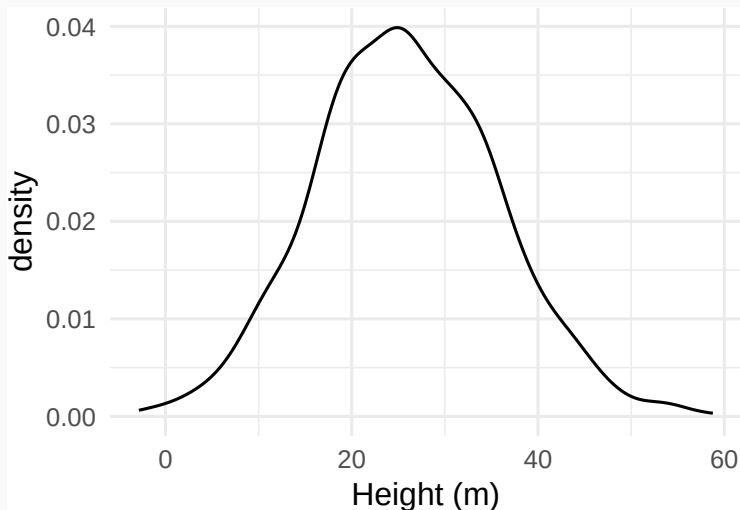
Reasonable prior



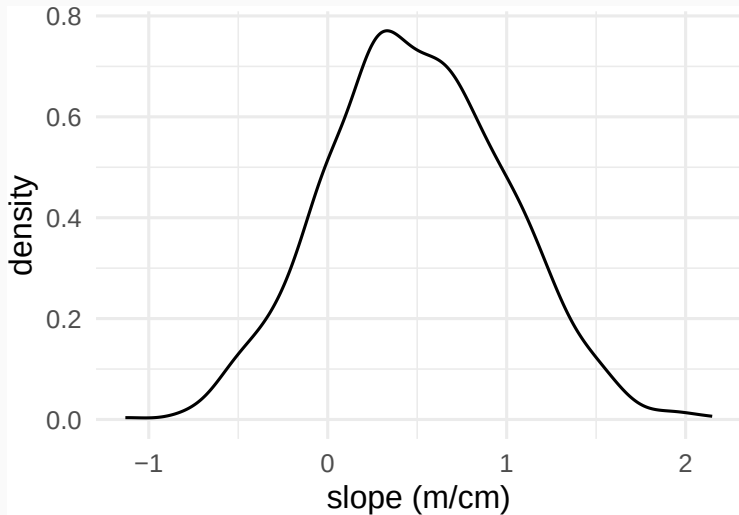
Defining priors for our trees example

```
priors <- c(  
  set_prior('normal(30, 10)', class = 'Intercept'),  
  set_prior('normal(0.5, 0.4)', class = 'b'),  
  set_prior('normal(0, 5)', class = 'sigma')  
)
```

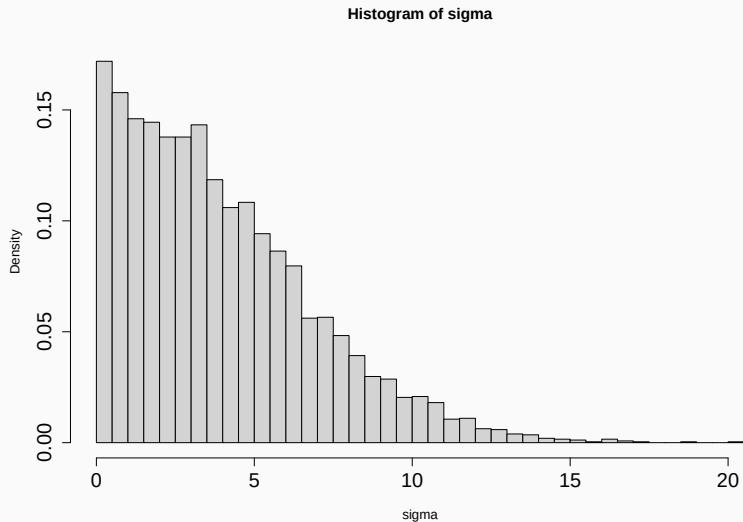
Prior for intercept (average height of 25-cm diameter tree)



Prior for slope



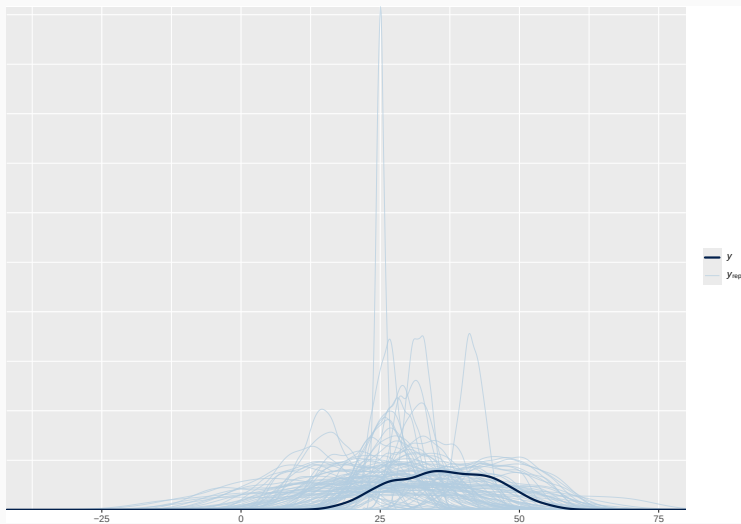
Prior for sigma (residual sd)



```
height.mod <- brm(height.formu,  
  data = trees,  
  prior = priors,  
  sample_prior = 'only')
```

Prior predictive check

```
pp_check(height.mod, ndraws = 100)
```



```
height.mod <- brm(height.formu,  
  data = trees,  
  prior = priors)
```


Model summary

```
summary(height.mod)
```

```
Family: gaussian
Links: mu = identity; sigma = identity
Formula: height ~ dbh.c
Data: trees (Number of observations: 1000)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	34.73	0.13	34.47	34.99	1.00	4222	3247
dbh.c	0.62	0.01	0.60	0.64	1.00	4499	2673

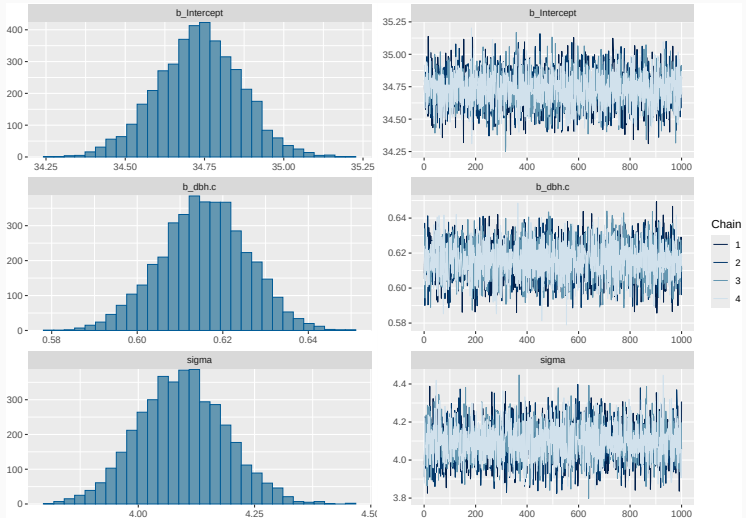
Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	4.09	0.09	3.92	4.28	1.00	4430	3153

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

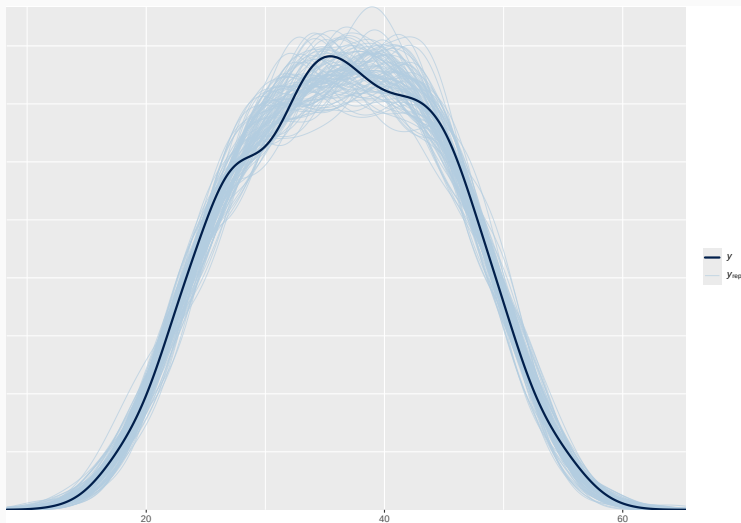
Model visualisation

```
plot(height.mod)
```



Posterior predictive checking

```
pp_check(height.mod, ndraws = 100)
```



```
library('shinystan')  
launch_shinystan(height.mod)
```

Sensitivity based on cjs_dist

Prior selection: all priors

Likelihood selection: all data

variable	prior	likelihood	diagnosis
b_Intercept	0.001	0.084	-
b_dbh.c	0.001	0.087	-
sigma	0.003	0.100	-
Intercept	0.001	0.085	-

Prior sensitivity analysis

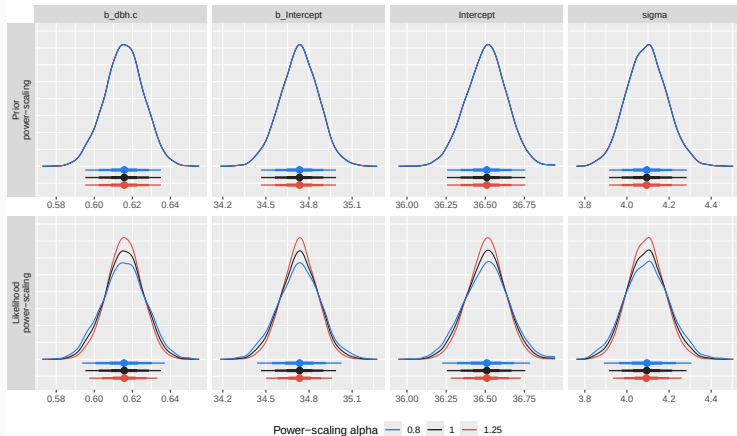
Power-scaling sensitivity

Posterior density estimates depending on amount of power-scaling (α).

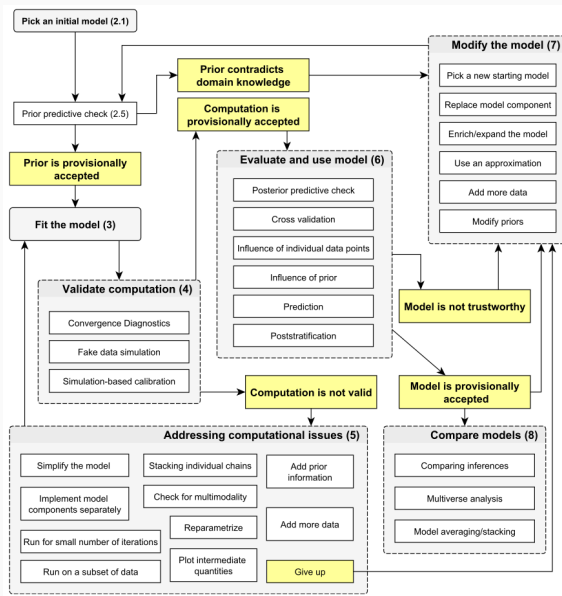
Overlapping lines indicate low sensitivity.

Wider gaps between lines indicate greater sensitivity.

Estimates with high Pareto k (dashed lines) may be inaccurate.



The Bayesian workflow



height ~ sex

[Regression and other stories](#)

[Statistical Rethinking](#)

[Statistical rethinking with brms, ggplot2, and the tidyverse](#)

[Bayesian Population Analysis using WinBugs](#)

[Applied Hierarchical Modeling in Ecology](#)

[Stan user guide](#)